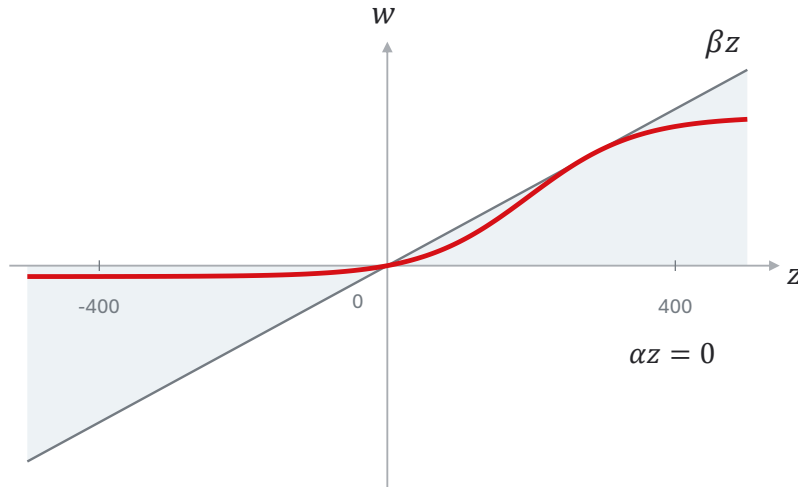
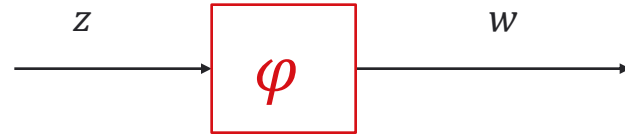


The shifted sigmoid lies in a sector



STN activation about its equilibrium



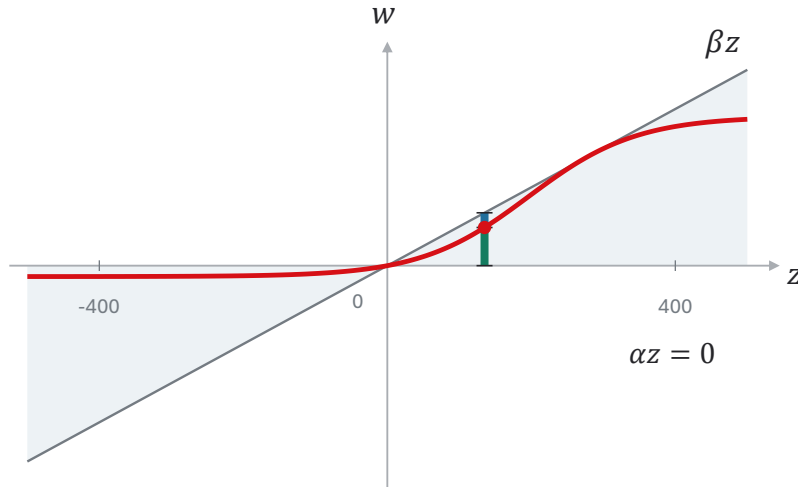
$$w = \varphi(z), \quad \varphi(0) = 0$$

$$\alpha = 0, \quad \beta \approx 0.7344$$

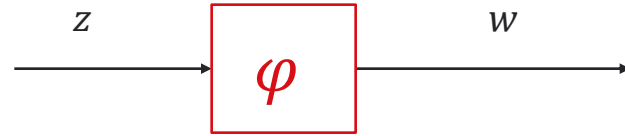
$$\varphi(z) = \frac{M}{2} \left[\tanh\left(\frac{2(z^* + z)}{M}\right) - \tanh\left(\frac{2z^*}{M}\right) \right]$$

$$M = 300, \quad z^* \approx -196.17$$

The sector gives two signed gaps



STN activation about its equilibrium



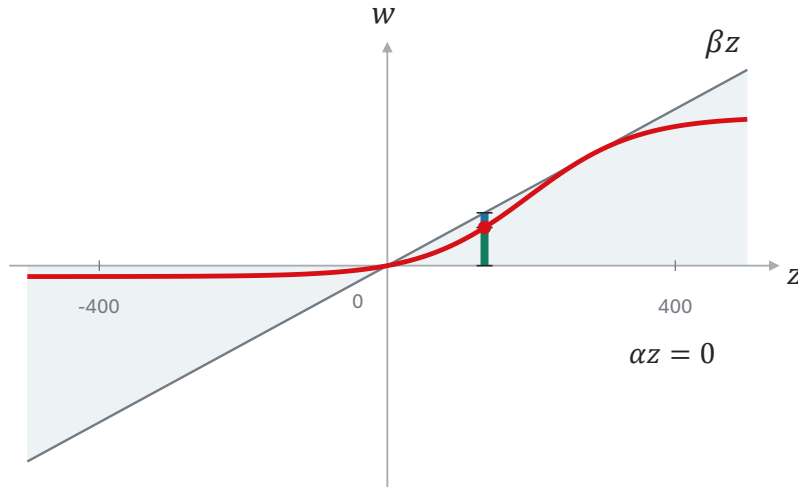
$$a = \beta z - w$$

$$b = w - \alpha z = w$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \beta & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z \\ w \end{bmatrix}$$

Inside the sector, the two gaps have the same sign.

Transform the gaps into input and output



STN activation about its equilibrium

$$\tilde{z} = a + b = \beta z$$

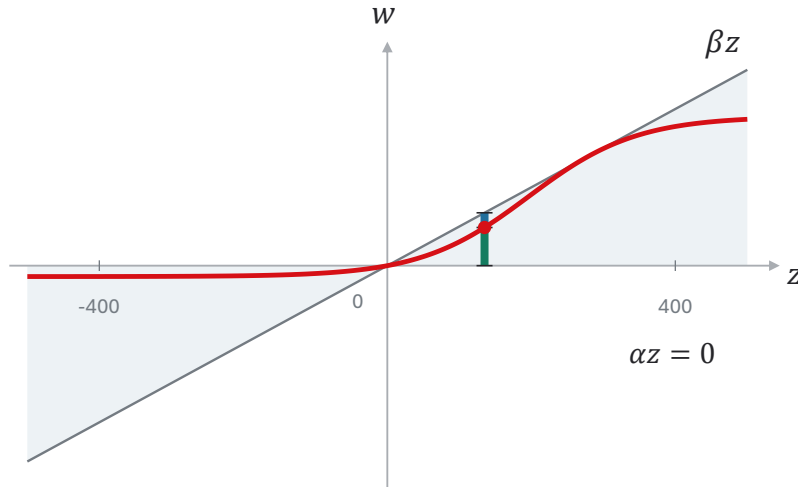
$$\tilde{w} = a - b = \beta z - 2w$$

$$|\tilde{w}| \leq |\tilde{z}|$$

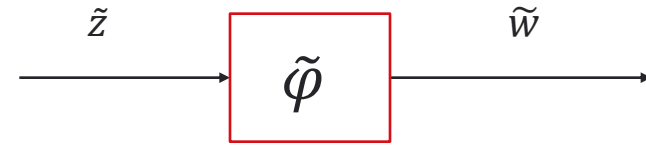
$$\begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z \\ w \end{bmatrix}$$

Choose unit scaling: the transformed input depends only on z .

The transformed nonlinearity is dissipative



STN activation about its equilibrium



Output energy cannot exceed input energy.

$$V \equiv 0$$

$$\sigma = \tilde{z}^2 - \tilde{w}^2 = 4ab \geq 0$$

$$V(T) - V(0) = 0 \leq \int_0^T \sigma(t) dt \quad \forall T \geq 0$$